

# AI YouTube Intelligence Dashboard

## Project Report

### 1. Introduction

The AI YouTube Intelligence Dashboard is an advanced business intelligence and analytics project developed using Python, YouTube Data API, and Microsoft Power BI.

The system extracts live YouTube channel and video data, processes it through Python-based ETL pipelines, and visualizes insights using interactive dashboards in Power BI.

The project aims to analyze:

- creator performance
- engagement metrics
- audience interaction
- content trends
- competitor intelligence

### 2. Objectives

The main objectives of the project are:

- To collect live YouTube analytics data
- To build an automated ETL pipeline
- To create interactive BI dashboards
- To analyze creator engagement patterns
- To compare channel performance
- To develop professional analytics reports

### 3. Technologies Used

Technology: Python

Purpose: Data extraction & processing

Technology: YouTube Data API v3

Purpose: Data source

Technology: Pandas

Purpose: Data manipulation

Technology: Power BI

Purpose: Dashboard visualization

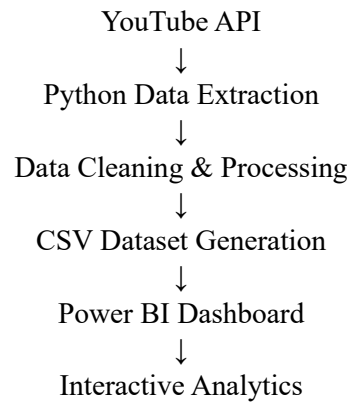
Technology: dotenv

Purpose: API security

Technology: CSV

Purpose: Data storage

#### 4. System Architecture



#### 5. Data Collected

- Channel Name
- Subscribers
- Total Views
- Total Videos
- Video Title
- Likes
- Comments
- Publish Date
- Engagement Rate

#### 6. Dashboard Modules

##### **Executive Overview**

- Total subscribers
- Total views
- Total videos
- Engagement metrics

##### **Video Performance Analytics**

- Most viewed videos
- Most liked videos
- Most commented videos

##### **Engagement Intelligence**

- Engagement trend analysis
- Audience interaction insights
- Engagement efficiency metrics

##### **Trend & Competitor Intelligence**

- Channel performance
- Audience reach
- Trend movement
- Growth analysis

## 7. KPI Calculations

### Engagement Rate

$(\text{Likes} + \text{Comments}) / \text{Views} \times 100$

### Views Per Video

$\text{Total Views} / \text{Number of Videos}$

### Engagement Efficiency

$\text{Total Interactions} / \text{Total Views}$

## 8. Features

- Live API-based data extraction
- Dynamic dashboard updates
- Professional UI design
- Interactive filters and slicers
- Comparative analytics
- KPI engineering

## 9. Future Enhancements

Future improvements may include:

- Real-time streaming analytics
- AI sentiment analysis
- Machine learning forecasting
- SQL database integration
- Recommendation systems

## 10. Conclusion

The AI YouTube Intelligence Dashboard successfully demonstrates the integration of APIs, ETL pipelines, Python data processing, and Power BI analytics into a unified business intelligence solution.

The project reflects modern analytics engineering practices and provides meaningful insights into YouTube creator performance and audience engagement.